ELECTRONIC PROTOTYPING AND AUTOMATION REPORT

1. Introduction

Electronic Prototyping and Automation is a field that involves designing, developing, testing, and controlling electronic systems to perform specific tasks automatically. During our studies, we learned about electronic circuit drawing, PCB design and formation, and the basic principles of automation.

One of the projects planned for practical implementation is a **Room Temperature Regulation System Using a Fan**. The system is intended to automatically control room temperature by switching the fan ON or OFF depending on the detected temperature.

At the time of writing this report, the required equipment and components have not yet been provided. Therefore, practical construction and testing of the project will be carried out after the equipment is made available.

2. Objectives

- To understand the principles of electronic circuit design.
- To learn how to draw electronic circuits.
- To understand the process of designing and forming a PCB.
- To learn how electronic components are connected in a circuit.
- To understand the principles of automation.
- To develop a temperature regulation system using a fan.
- To gain practical skills in designing and testing electronic systems.

3. Electronic Circuit Drawing

Electronic circuit drawing involves representing electronic components and their connections using standard symbols. It helps engineers and students understand how a circuit is connected before physically assembling it. Circuit diagrams provide a clear guide for assembling and troubleshooting electronic systems.

4. PCB Design and Formation

A Printed Circuit Board (PCB) is a board used to mechanically support and electrically connect electronic components. Instead of connecting components using many separate wires, conductive tracks are created on the PCB to provide electrical connections.

The general PCB development process includes: drawing the circuit schematic, selecting components, designing the PCB layout, placing components, routing electrical tracks, checking the design, producing the PCB, mounting components, and testing the completed circuit.

5. Proposed Temperature Regulation Project

The proposed project is a **Room Temperature Regulation System Using a Fan**. The purpose of the system is to monitor the temperature inside a room and automatically control a fan. When the temperature rises above a predetermined level, the fan will be switched ON. When the temperature falls to an acceptable level, the fan will be switched OFF.

6. Expected Components

- Temperature sensor
- Microcontroller or control circuit
- DC fan
- Relay or transistor
- Resistors
- Connecting wires
- Power supply
- PCB
- LED indicators
- Breadboard for prototyping

7. Proposed Operation

Temperature Sensor → **Control Unit** → **Switching Device** → **Fan**

The temperature sensor will detect the temperature of the room. The control unit will process the temperature information and determine whether the fan should operate. If the temperature reaches a predetermined value, the control unit will activate the switching device and the fan will turn ON. When the temperature decreases, the fan will turn OFF.

8. Expected Results

- The system should detect room temperature.
- The fan should turn ON when the temperature becomes high.
- The fan should turn OFF when the temperature decreases.
- The system should operate automatically.
- The circuit should function correctly after testing.

9. Challenges

Since the practical implementation has not yet started, the main current challenge is the unavailability of the required equipment and components. Other challenges may be identified during the construction and testing stages.

10. Conclusion

Electronic Prototyping and Automation provides important knowledge and practical skills in electronic circuit design, PCB development, and automation. Through this course, we have learned the basic principles of drawing electronic circuits and designing PCBs.

The planned room temperature regulation project will provide practical experience in applying these concepts to a real-world automation system. The practical construction, testing, and evaluation of the project will be conducted once the required equipment and components are provided.